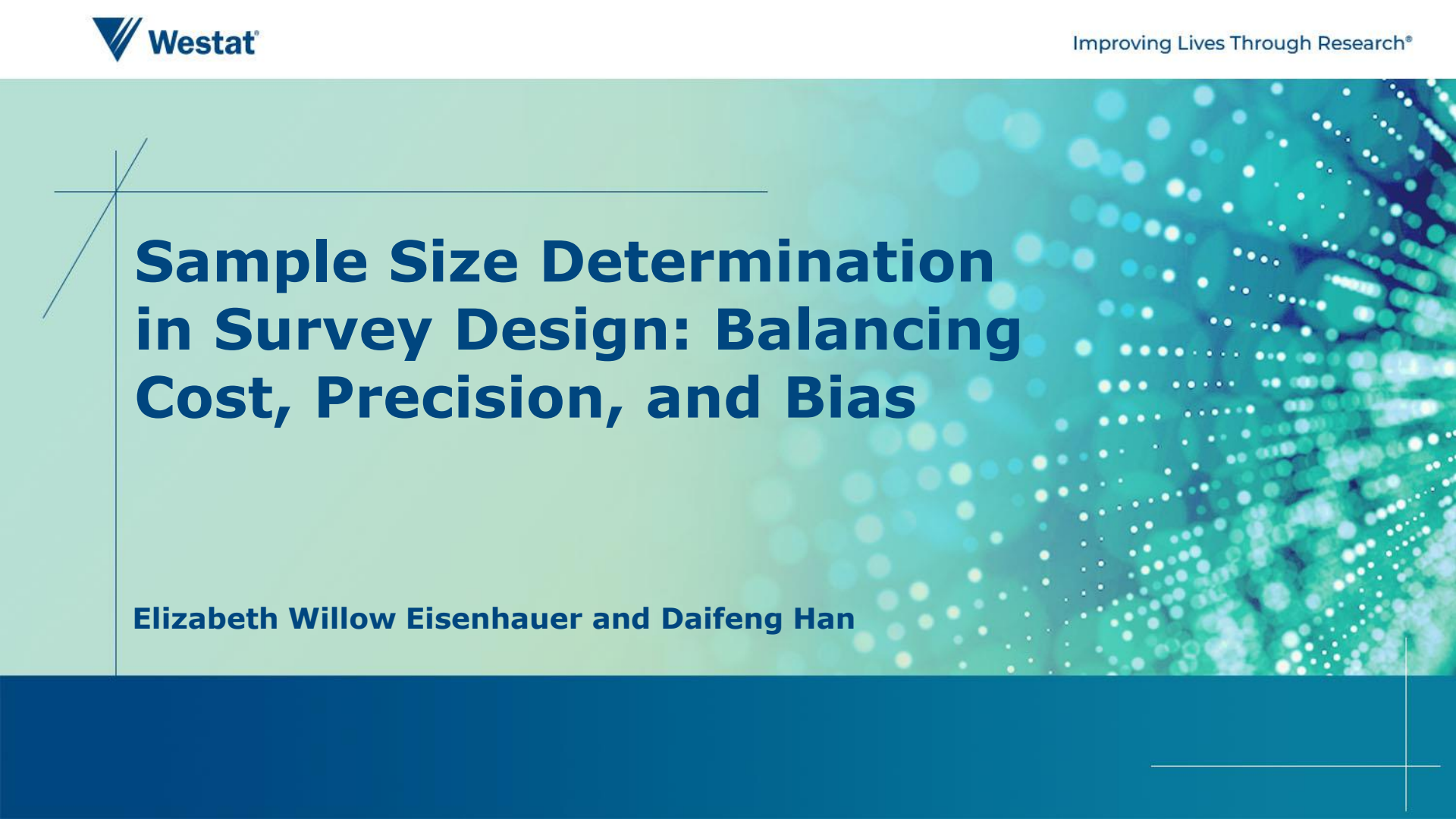




Sample Size Determination in Survey Design: Balancing Cost, Precision, and Bias

Elizabeth Willow Eisenhauer and Daifeng Han

Cost impacts the variance and bias of estimates.

Given a budget constraint,

- Optimize sample size and allocation
- Increase precision
- Reduce bias risk

The goal is to allocate cost where it most improves inferential power for key outcomes (**value**).

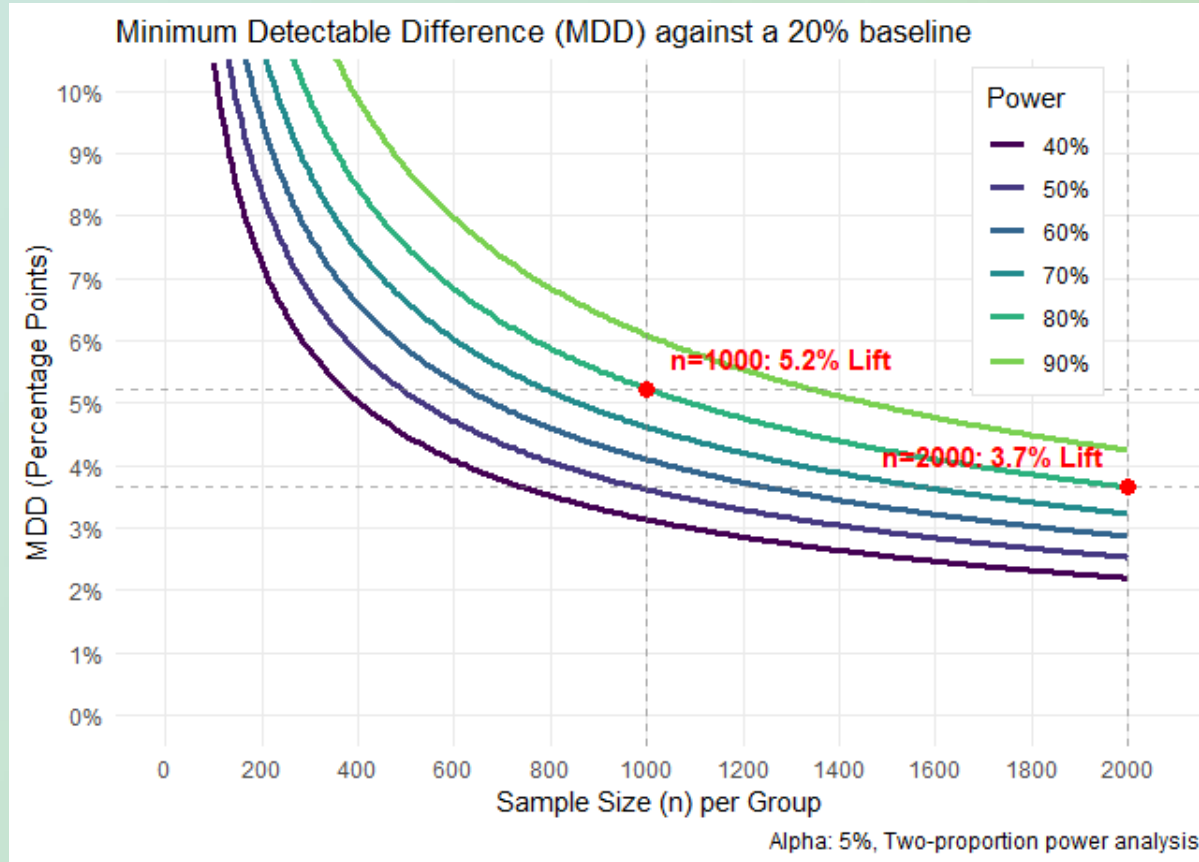
What are the key drivers of survey cost?

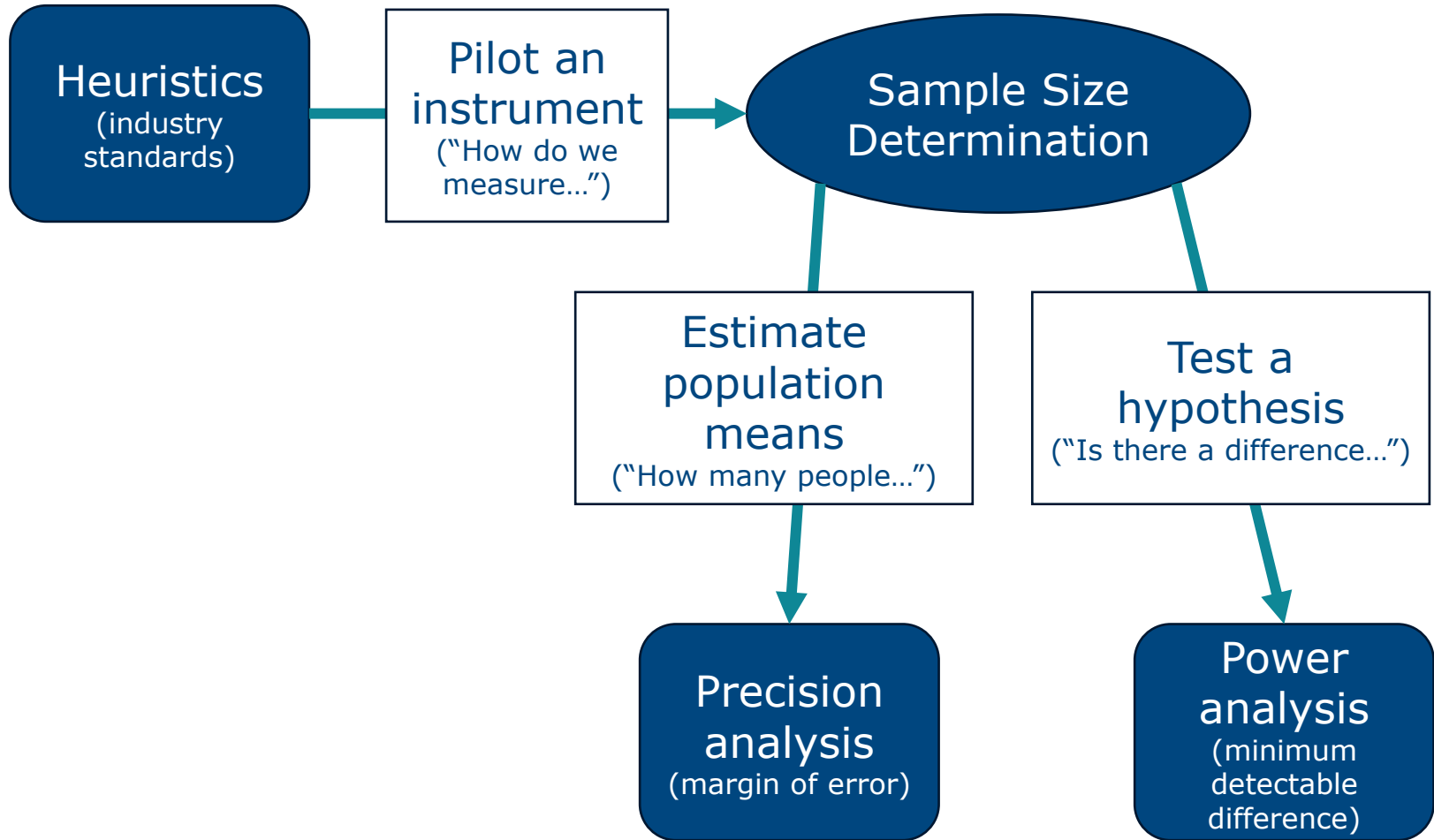
- Sample size and allocation across groups
- Expected response rates and follow-up intensity
- Mode (web, phone, in-person)
- Clustering and field logistics
- Length and complexity of the instrument

These same levers also drive variance and bias.

We need modeling tools that connect these levers.

Design levers are **implicit assumptions** in power analysis.





Power and precision analysis as a communication tool

Sample size calculations make cost-error tradeoffs explicit by **formalizing the relationship** between:

- Sample size
- Outcome prevalence / variance
- Precision (CI width) or detectability (MDE / power)

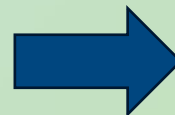
Bias is not part of the calculation... sensitivity analysis can help

Results communicate the impact of costs on estimates.

- What changes when we **allocate costs differently**?

Design question: How to allocate sample for comparing groups with different costs?

Allocation to Groups 1 & 2	Group 1 Sample Size	Group 2 Sample Size	Total Sample Size
Option A: 50/50	936	936	1,866
Option B: 33/66	696	1,416	2,106



B is **lower cost** with **the same statistical power**

- **Group 1 unit is 5x the cost of group 2 unit**
- **Option A: smaller sample size, higher cost**
- **Option B: larger sample size, lower cost**

Assumptions

- Significance: 5%
- Minimum Detectable Effect Size (MDES): 0.15
- Power: 80%

In the design phase, we discuss

1. What is the **design goal**, and what do we expect?
2. What does **data collection** look like?
3. What does nonresponse **follow-up** look like?
4. What does **data processing and analysis** look like?

To balance cost, precision, and bias.

1. What is the design goal, and what do we expect?

- Competing **goals**
 - **Overall** estimates
 - vs. **domain** estimates (larger sample size required; higher cost)
 - vs. domain **differences** (even larger sample size & cost)
- What do we expect to see in this **population**?
 - Effect sizes or subgroup difference expectations need empirical **benchmarking** (prior studies or a pilot study)
 - Are **auxiliary** data available to support the sample design?
 - Some **measures** are more sensitive to change than others

2. What does data collection look like?

- The **design effect** quantifies how the effective sample size differs from the nominal sample size
 - **Oversampling** some domains improves domain estimates while increasing the design effect (meaning you need a larger overall sample for the same precision)
- A **less-expensive mode** can increase overall sample size while decreasing response rates
- **Nonresponse behavior** may differ by domain (e.g., region) and mode
 - **Responsive design** (and **auxiliary data**) can help address issues during data collection
 - "If nonrespondents and respondents differ by X" **sensitivity analysis**
- **Study horizon** impacts the cost function. Higher proportion of cost is fixed cost for longitudinal studies

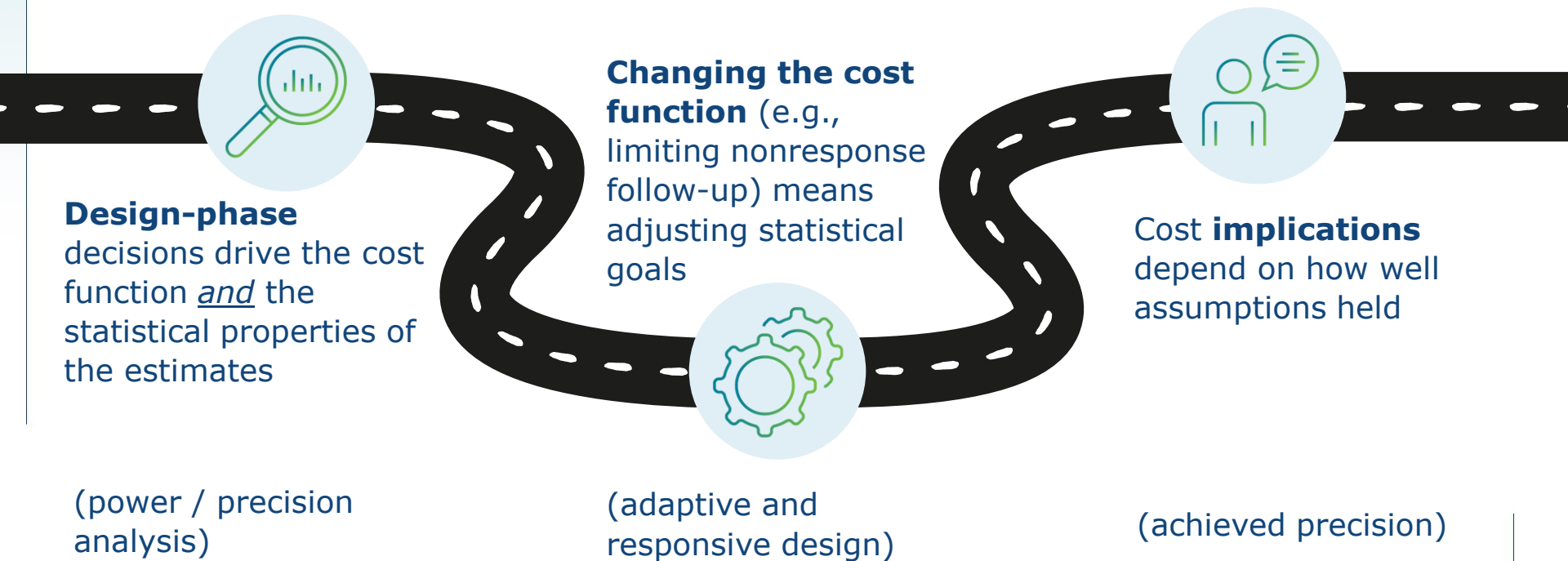
3. What does nonresponse follow-up look like?

- Nonresponse follow-up **decreases nonresponse bias** but is often more **costly**
 - Different mode at follow-up than at primary data collection introduces **mode effect**
- **Targeted follow-up** can more effectively reduce bias while increasing weight variation
 - For example, in-person follow-up requires a **clustered** sample, but the clustering effect decreases the effective sample size

4. What does data processing and analysis look like?

- Using auxiliary data (from prior waves or external vendors) to **impute missing data** can increase effective sample size and reduce costly follow-up.
- With **strong covariates**, we may not need as large a sample for the same precision.
- Conditioning on auxiliary variables through weighting or in the analysis can change the **precision and bias tradeoff**.

Sample size determination is not just for design phase



Sample Size Conversation Starters

In initial planning:

- What **decisions** are fixed, and where would it be helpful to see design **options**?

During re-assessment of assumptions mid-study:

- Which initial design assumptions have been confirmed or challenged by **field experience**?

During interpretation:

- What **conclusions** are warranted given the study's precision, limitations, and scope?

Thank you

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